Operations Research 1: Deterministic Models
Integer Programming: Introduction and Modeling

Dr. Lei Zhao

Department of Industrial Engineering

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Operations & Services Research (TOpS) Laboratory
Department of Industrial Engineering
Tsinghua University, China

Wyndor Example (Revisit)

Wyndor Glass Co.

- Two products
  - Product 1
    - Glass door w/ aluminum framing
  - Product 2
    - Wood-frame window
- Three plants
  - Plant 1
    - Aluminum frames and hardware
  - Plant 2
    - Wood frames
  - Plant 3
    - Glass and product assemble
- Data
  - Available production time in each plant (hours / week)
  - Production time used in each plant for each batch of each product (hours / batch)
  - Profit per batch of each product

	Doors	Windows		
Profit per batch	$3,000	$5,000		
				Hours
	Hours used per batch produced			available
Plant 1	1	0		4
Plant 2	0	2		12
Plant 3	3	2		18

- Decision: production
- Objective: to maximize the total profit (per week)

Wyndor Glass Co. (Revisit)

TOpS

- Mathematical formulation
  - x_1 = # batches of product 1 produced per week
  - x_2 = # batches of product 2 produced per week
  - z = total profit per week

	Doors	Windows			
Profit per batch	$3,000	$5,000			
					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

$$
\begin{aligned}
\max \quad & z = 3x_1 + 5x_2 \\
\text{s.t.} \quad & x_1 \leq 4, \\
& 2x_2 \leq 12, \\
& 3x_1 + 2x_2 \leq 18, \\
& x_1, \quad x_2 \geq 0.
\end{aligned}
$$

	Doors	Windows			
Profit per batch	$3,000	$5,000			
			Hours		Hours
	Hours used per batch produced		used		available
Plant 1	1	0	2	<=	4
Plant 2	0	2	12	<=	12
Plant 3	3	2	18	<=	18
	Doors	Windows			Total profit
Batches produced	2	6			$36,000

Wyndor Glass Co.

- Mathematical formulation
  - x_1 = # batches of product 1 produced per week
  - x_2 = # batches of product 2 produced per week
  - z = total profit per week

	Doors	Windows			
Profit per batch	$3,000	$5,000			
					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2.5			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

$$
\begin{aligned}
\max \quad & z = 3x_1 + 5x_2 \\
\text{s.t.} \quad & x_1 \leq 4, \\
& 2.5x_2 \leq 12, \\
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					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2.5			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

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& 2.5x_2 \le 12, \\
& 3x_1 + 2x_2 \le 18, \\
& x_1, \quad x_2 \ge 0.
\end{aligned}
$$

	Doors	Windows			
Profit per batch	$3,000	$5,000			
			Hours		Hours
	Hours used per batch produced		used		available
Plant 1	1	0	2.8	<=	4
Plant 2	0	2.5	12	<=	12
Plant 3	3	2	18	<=	18
	Doors	Windows			Total profit
Batches produced	2.8	4.8			$32,400

Wyndor Glass Co.

- Mathematical formulation
  - x_1 = # batches of product 1 produced per week
  - x_2 = # batches of product 2 produced per week
  - z = total profit per week

	Doors	Windows			
Profit per batch	$3,000	$5,000			
					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2.5			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

$$
\begin{aligned}
\max \quad & z = 3x_1 + 5x_2 \\
\text{s.t.} \quad & x_1 \le 4, \\
& 2.5x_2 \le 12, \\
& 3x_1 + 2x_2 \le 18, \\
& x_1, \quad x_2 \ge 0 \text{ and integer.}
\end{aligned}
$$

*Wyndor Glass Co.

- Mathematical formulation
  - x_1 = # batches of product 1 produced per week
  - x_2 = # batches of product 2 produced per week
  - z = total profit per week

	Doors	Windows			
Profit per batch	$3,000	$5,000			
					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2.5			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

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\begin{aligned}
\max \quad & z = 3x_1 + 5x_2 \\
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	Doors	Windows			
Profit per batch	$3,000	$5,000			
					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2.5			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

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\end{aligned}
$$

	Doors	Windows			
Profit per batch	$3,000	$5,000			
			Hours		Hours
	Hours used per batch produced		used		available
Plant 1	1	0	3	<=	4
Plant 2	0	2.5	10	<=	12
Plant 3	3	2	17	<=	18
	Doors	Windows			Total profit
Batches produced	3	4			$29,000

Four Assumptions of LP (Revisit)

- Proportionality
  - Objective value/LHS is proportional to the level of activity
  - Nonlinear/mixed integer programming
- Additivity
  - Sum of individual contributions (objective/LHS)
  - Nonlinear programming
- Divisibility
  - (Mixed) Integer programming
- Certainty
  - Sensitivity analysis, but …
  - Stochastic programming, (approximate) dynamic programming, simulation optimization, robust optimization, …

More Modeling Examples

*Optimization Problem/Model: Key Elements

- 我们知道什么? What do we know?

  流程 + 数据 process + data

- 我们需要决定什么？ What do we need to decide?

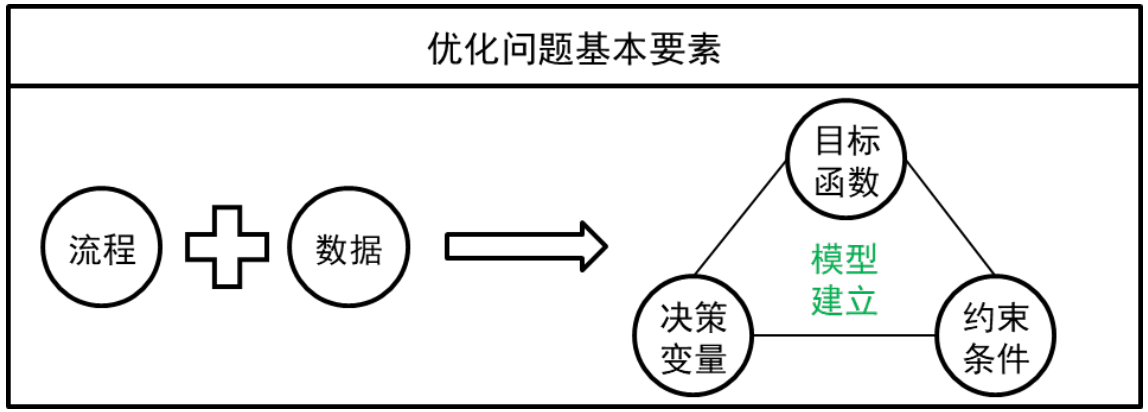
  决策变量 decision variables

- 我们希望达到什么目标？ What do we want to achieve?

  目标函数 objective function

- 我们的决策受到什么限制？ What prohibits our ability to do so?

  约束条件 constraints



Example: Investments at Tatham

Investments at Tatham

- Seven investments
  - Cash required and NPV
- Total budget: $15,000
  - Remaining cash cannot be kept.
- Objective: maximize total NPV
- What are our decisions?

Table 6.1 Data for the Capital Budgeting Example

Investment	Cash Required	NPV
1	$5,000	$16,000
2	$2,500	$8,000
3	$3,500	$10,000
4	$6,000	$19,500
5	$7,000	$22,000
6	$4,500	$12,000
7	$3,000	$7,500

*Investments at Tatham

- Seven investments
  - Cash required and NPV
- Total budget: $15,000
  - Remaining cash cannot be kept.
- Objective: maximize total NPV
- What are our decisions?
  - Whether to invest in investment i, $i = 1, \ldots, 7$

Table 6.1 Data for the Capital Budgeting Example

Investment	Cash Required	NPV
1	$5,000	$16,000
2	$2,500	$8,000
3	$3,500	$10,000
4	$6,000	$19,500
5	$7,000	$22,000
6	$4,500	$12,000
7	$3,000	$7,500

Investments at Tatham

- Data
- Decisions
- Model

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Investment	Cash Required	NPV
1	$5,000	$16,000
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4	$6,000	$19,500
5	$7,000	$22,000
6	$4,500	$12,000
7	$3,000	$7,500

*Investments at Tatham

- Data

$$c_i = \text{cash required of investment } i \in I$$
$$N_i = \text{NPV of investment } i \in I$$
$$B = \text{budget}$$

- Decisions

- Model

Table 6.1 Data for the Capital Budgeting Example

Investment	Cash Required	NPV
1	$5,000	$16,000
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5	$7,000	$22,000
6	$4,500	$12,000
7	$3,000	$7,500

*Investments at Tatham

- Data

$$c_i = \text{cash required of investment } i \in I$$
$$N_i = \text{NPV of investment } i \in I$$
$$B = \text{budget}$$

- Decisions

$$x_i = \begin{cases} 1, & \text{if we invest in investment } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

- Model

Table 6.1 Data for the Capital Budgeting Example

Investment	Cash Required	NPV
1	$5,000	$16,000
2	$2,500	$8,000
3	$3,500	$10,000
4	$6,000	$19,500
5	$7,000	$22,000
6	$4,500	$12,000
7	$3,000	$7,500

*Investments at Tatham

Data

$$c_i = \text{cash required of investment } i \in I$$
$$N_i = \text{NPV of investment } i \in I$$
$$B = \text{budget}$$

Table 6.1 Data for the Capital Budgeting Example

Investment	Cash Required	NPV
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4	$6,000	$19,500
5	$7,000	$22,000
6	$4,500	$12,000
7	$3,000	$7,500

Decisions

$$x_i = \begin{cases} 1, & \text{if we invest in investment } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

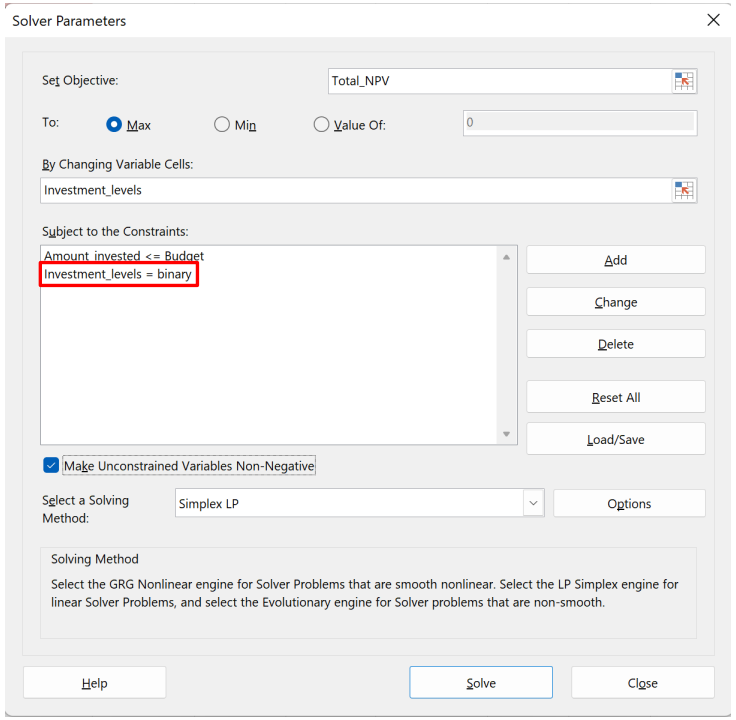
Model

$$\begin{array}{lll} \max & \sum_{i \in I} N_i x_i & \text{maximize the total NPV} \\ \text{s.t.} & \sum_{i \in I} c_i x_i \le B & \text{budget constraint} \\ & x_i \text{ binary}, \forall i \in I & \text{equivalently, } x_i \in \{0,1\}, \forall i \in I \end{array}$$

*Investments at Tatham

Input data on potential investments							
Investment	1	2	3	4	5	6	7
Investment cost	$5,000	$2,500	$3,500	$6,000	$7,000	$4,500	$3,000
NPV	$16,000	$8,000	$10,000	$19,500	$22,000	$12,000	$7,500
NPV per investment dollar							
Decisions: whether to invest							
Investment levels							
Budget constraint							
	Amount invested		Budget				
			$15,000				
Objective to maximize							
Total NPV							

Input data on potential investments							
Investment	1	2	3	4	5	6	7
Investment cost	$5,000	$2,500	$3,500	$6,000	$7,000	$4,500	$3,000
NPV	$16,000	$8,000	$10,000	$19,500	$22,000	$12,000	$7,500
NPV per investment dollar	3.20	3.20	2.86	3.25	3.14	2.67	2.50
Decisions: whether to invest							
Investment levels	1	1	0	0	1	0	0
Budget constraint							
	Amount invested		Budget				
	$14,500	<=	$15,000				
Objective to maximize							
Total NPV	$46,000						



*Investments at Tatham

- Data

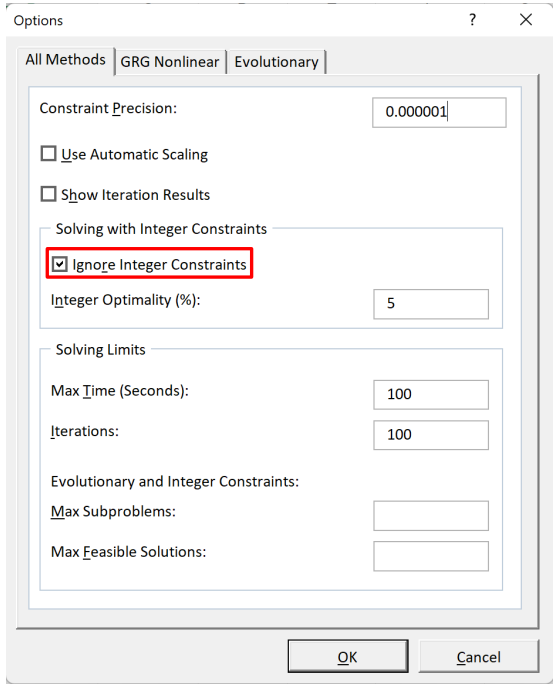
$$c_i = \text{cash required of investment } i \in I$$
$$N_i = \text{NPV of investment } i \in I$$
$$B = \text{budget}$$

- Decisions

$$x_i = \begin{cases} 1, & \text{if we invest in investment } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

- Model (relaxed to LP)

$$\begin{array}{lll} \max & \sum_{i \in I} N_i x_i & \text{maximize the total NPV} \\ \text{s.t.} & \sum_{i \in I} c_i x_i \leq B & \text{budget constraint} \\ & 0 \leq x_i \leq 1, \forall i \in I & \end{array}$$



*Investments at Tatham

Not keeping remaining cash

$$
\begin{array}{lll}
\max & \sum_{i \in I} N_i x_i & \text{maximize the total NPV} \\
\text{s.t.} & \sum_{i \in I} c_i x_i \leq B & \text{budget constraint} \\
 & x_i \text{ binary}, \forall i \in I & \text{equivalently, } x_i \in \{0,1\}, \forall i \in I
\end{array}
$$

Input data on potential investments							
Investment	1	2	3	4	5	6	7
Investment cost	$5,000	$2,500	$3,500	$6,000	$7,000	$4,500	$3,000
NPV	$16,000	$8,000	$10,000	$19,500	$22,000	$12,000	$7,500
NPV per investment dollar	3.20	3.20	2.86	3.25	3.14	2.67	2.50
Decisions: whether to invest							
Investment levels	1	1	0	0	1	0	0
Budget constraint							
	Amount invested		Budget				
	$14,500	<=	$15,000				
Objective to maximize							
Total NPV	$46,000						

Relaxation on x binary

$$
\begin{array}{lll}
\max & \sum_{i \in I} N_i x_i & \text{maximize the total NPV} \\
\text{s.t.} & \sum_{i \in I} c_i x_i \leq B & \text{budget constraint} \\
 & 0 \leq x_i \leq 1, \forall i \in I &
\end{array}
$$

Input data on potential investments							
Investment	1	2	3	4	5	6	7
Investment cost	$5,000	$2,500	$3,500	$6,000	$7,000	$4,500	$3,000
NPV	$16,000	$8,000	$10,000	$19,500	$22,000	$12,000	$7,500
NPV per investment dollar	3.20	3.20	2.86	3.25	3.14	2.67	2.50
Decisions: whether to invest							
Investment levels	1	1	0	1	0.21	0	0
Budget constraint							
	Amount invested		Budget				
	$15,000	<=	$15,000				
Objective to maximize							
Total NPV	$48,214						

Investments at Tatham

- What if you can keep the remaining cash?

*Investments at Tatham

- Data

$$c_i = \text{cash required of investment } i \in I$$
$$N_i = \text{NPV of investment } i \in I$$
$$B = \text{budget}$$

- Decisions

- Model

Table 6.1 Data for the Capital Budgeting Example

Investment	Cash Required	NPV
1	$5,000	$16,000
2	$2,500	$8,000
3	$3,500	$10,000
4	$6,000	$19,500
5	$7,000	$22,000
6	$4,500	$12,000
7	$3,000	$7,500

*Investments at Tatham

- Data

$$c_i = \text{cash required of investment } i \in I$$
$$N_i = \text{NPV of investment } i \in I$$
$$B = \text{budget}$$

- Decisions

$$x_i = \begin{cases} 1, & \text{if we invest on investment } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$
$$S = \text{remaining (uninvested) cash.}$$

- Model

Table 6.1 Data for the Capital Budgeting Example

Investment	Cash Required	NPV
1	$5,000	$16,000
2	$2,500	$8,000
3	$3,500	$10,000
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*Investments at Tatham

Data

$$c_i = \text{cash required of investment } i \in I$$
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Investment	Cash Required	NPV
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4	\$6,000	\$19,500
5	\$7,000	\$22,000
6	\$4,500	\$12,000
7	\$3,000	\$7,500

Decisions

$$x_i = \begin{cases} 1, & \text{if we invest on investment } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

$$S = \text{remaining (uninvested) cash.}$$

Model

$$\begin{array}{lll} \max & \sum_{i \in I} N_i x_i + S & \text{maximize total NPV plus remaining cash} \\ \text{s.t.} & \sum_{i \in I} c_i x_i + S = B & \text{budget constraint} \\ & x_i \text{ binary}, \forall i \in I & \text{equivalently, } x_i \in \{0,1\}, \forall i \in I \\ & S \geq 0. & \end{array}$$

*Investments at Tatham

Not keeping remaining cash

$$
\begin{array}{lll}
\max & \sum_{i \in I} N_i x_i & \text{maximize the total NPV} \\
\text{s.t.} & \sum_{i \in I} c_i x_i \leq B & \text{budget constraint} \\
 & x_i \text{ binary}, \forall i \in I & \text{equivalently, } x_i \in \{0,1\}, \forall i \in I
\end{array}
$$

Input data on potential investments							
Investment	1	2	3	4	5	6	7
Investment cost	$5,000	$2,500	$3,500	$6,000	$7,000	$4,500	$3,000
NPV	$16,000	$8,000	$10,000	$19,500	$22,000	$12,000	$7,500
NPV per investment dollar	3.20	3.20	2.86	3.25	3.14	2.67	2.50
Decisions: whether to invest							
Investment levels	1	1	0	0	1	0	0
Budget constraint							
	Amount invested		Budget				
	$14,500	<=	$15,000				
Objective to maximize							
Total NPV	$46,000						

Keeping remaining cash

$$
\begin{array}{lll}
\max & \sum_{i \in I} N_i x_i + S & \text{maximize total NPV plus remaining cash} \\
\text{s.t.} & \sum_{i \in I} c_i x_i + S = B & \text{budget constraint} \\
 & x_i \text{ binary}, \forall i \in I & \text{equivalently, } x_i \in \{0,1\}, \forall i \in I \\
 & S \geq 0. &
\end{array}
$$

Input data on potential investments							
Investment	1	2	3	4	5	6	7
Investment cost	$5,000	$2,500	$3,500	$6,000	$7,000	$4,500	$3,000
NPV	$16,000	$8,000	$10,000	$19,500	$22,000	$12,000	$7,500
NPV per investment dollar	3.20	3.20	2.86	3.25	3.14	2.67	2.50
Decisions: whether to invest							
Investment levels	1	1	0	0	1	0	0
Budget constraint							
	Amount invested		Budget				
	$14,500		$15,000				
			=				
	Remaining		Total cash				
	$500		$15,000				
Objective to maximize							
Total NPV	$46,000						
Total remaining	$500						
Total	$46,500						

Example: Textile Manufacturing

Textile Manufacturing

- Products: shirts, shorts, pants, skirts, and jackets
  - Unit labor and cloth usage
  - Unit variable cost and unit sales price
  - Machinery with rental cost (weekly)

Table 6.3 Data for the Great Threads Example

	Rental Cost	Labor Hours	Cloth (sq yd)	Sales Price	Unit Variable Cost
Shirts	$1500	2.0	3.0	$35	$20
Shorts	$1200	1.0	2.5	$40	$10
Pants	$1600	6.0	4.0	$65	$25
Skirts	$1500	4.0	4.5	$70	$30
Jackets	$1600	8.0	5.5	$110	$35

- Total: 4000 hours & 4500 square yards of cloth (weekly)
- Objective: maximize total profit (weekly)

- What are our decisions?

Textile Manufacturing

- What are our decisions?

*Textile Manufacturing

- What are our decisions?

- Whether to produce any of each clothing
  - Binary
- How much (many) of each clothing to produce (weekly)
  - Continuous (integral)

Textile Manufacturing

- Data
- Decisions
- Model

Table 6.3 Data for the Great Threads Example

	Rental Cost	Labor Hours	Cloth (sq yd)	Sales Price	Unit Variable Cost
Shirts	$1500	2.0	3.0	$35	$20
Shorts	$1200	1.0	2.5	$40	$10
Pants	$1600	6.0	4.0	$65	$25
Skirts	$1500	4.0	4.5	$70	$30
Jackets	$1600	8.0	5.5	$110	$35

*Textile Manufacturing

- Data

f_i = fixed cost of clothing $i \in I$ $\quad s_i$ = cloth consumption of clothing $i \in I$

l_i = labor hours of clothing $i \in I$ $\quad L$ = total labor hours

p_i = unit price of clothing $i \in I$ $\quad S$ = total clothes

c_i = variable cost of clothing $i \in I$

- Decisions

- Model

Table 6.3 Data for the Great Threads Example

	Rental Cost	Labor Hours	Cloth (sq yd)	Sales Price	Unit Variable Cost
Shirts	$1500	2.0	3.0	$35	$20
Shorts	$1200	1.0	2.5	$40	$10
Pants	$1600	6.0	4.0	$65	$25
Skirts	$1500	4.0	4.5	$70	$30
Jackets	$1600	8.0	5.5	$110	$35

*Textile Manufacturing

- Data

f_i = fixed cost of clothing $i \in I$ $\quad s_i$ = cloth consumption of clothing $i \in I$

l_i = labor hours of clothing $i \in I$ $\quad L$ = total labor hours

p_i = unit price of clothing $i \in I$ $\quad S$ = total clothes

c_i = variable cost of clothing $i \in I$

- Decisions

$$x_i = \begin{cases} 1, & \text{if we produce clothing } i \in I \text{ / rent machinery for clothing } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

y_i = amount of clothing $i \in I$ to be produced

- Model

*Textile Manufacturing

- Data

f_i = fixed cost of clothing $i \in I$ $\quad s_i$ = cloth consumption of clothing $i \in I$

l_i = labor hours of clothing $i \in I$ $\quad L$ = total labor hours

p_i = unit price of clothing $i \in I$ $\quad S$ = total clothes

c_i = variable cost of clothing $i \in I$

- Decisions

$$x_i = \begin{cases} 1, & \text{if we produce clothing } i \in I \text{ / rent machinery for clothing } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

y_i = amount of clothing $i \in I$ to be produced

- Model

$$\begin{array}{lll} \max & \sum_{i \in I} p_i y_i - \sum_{i \in I} c_i y_i - \sum_{i \in I} f_i x_i & \text{profit} = \text{revenue} - \text{cost} \\ \text{s.t.} & \sum_{i \in I} l_i y_i \le L & \text{labor hours constraint} \\ & \sum_{i \in I} s_i y_i \le S & \text{cloth constraint} \\ & y_i \le M_i x_i, \quad \forall i \in I & \text{where } M_i = \min\{L/l_i, S/s_i\} \\ & x_i \text{ binary}, \quad \forall i \in I & \\ & y_i \ge 0, \quad \forall i \in I & \end{array}$$

*Textile Manufacturing

Production (continuous)

Input data on products					
	Shirts	Shorts	Pants	Skirts	Jackets
Labor hours	2	1	6	4	8
Cloth	3	2.5	4	4.5	5.5
Selling price	$35	$40	$65	$70	$110
Variable cost	$20	$10	$25	$30	$35
Fixed cost for equipment	$1,500	$1,200	$1,600	$1,500	$1,600
Production plan, constraints on capacity					
	Shirts	Shorts	Pants	Skirts	Jackets
Produce any?	0	1	0	0	1
Units produced	0.00	965.52	0.00	0.00	379.31
	<=	<=	<=	<=	<=
Logical upper limit	0.00	1800.00	0.00	0.00	500.00
Constraints on resources					
	Resource used		Available		
Labor hours	4000.00	<=	4000		
Cloth	4500.00	<=	4500		
Monetary outputs					
Revenue	$80,345				
Total variable cost	$22,931				
Total fixed cost for equipment	$2,800				
Profit	$54,614				

$$
\begin{array}{lll}
\max & \sum_{i\in I} p_i y_i - \sum_{i\in I} c_i y_i - \sum_{i\in I} f_i x_i & \text{profit} = \text{revenue} - \text{cost} \\
\text{s.t.} & \sum_{i\in I} l_i y_i \le L & \text{labor hours constraint} \\
 & \sum_{i\in I} s_i y_i \le S & \text{cloth constraint} \\
 & y_i \le M_i x_i, \quad \forall i \in I & \text{where } M_i = \min\{L/l_i, S/s_i\} \\
 & x_i \text{ binary}, \quad \forall i \in I & \\
 & y_i \ge 0, \quad \forall i \in I &
\end{array}
$$

*Textile Manufacturing

- Data

f_i = fixed cost of clothing $i \in I$ $\quad s_i$ = cloth consumption of clothing $i \in I$

l_i = labor hours of clothing $i \in I$ $\quad L$ = total labor hours

p_i = unit price of clothing $i \in I$ $\quad S$ = total clothes

c_i = variable cost of clothing $i \in I$

- Decisions

$$x_i = \begin{cases} 1, & \text{if we produce clothing } i \in I \text{ / rent machinery for clothing } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

y_i = amount of clothing $i \in I$ to be produced

- Model

$$\begin{array}{lll} \max & \sum_{i \in I} p_i y_i - \sum_{i \in I} c_i y_i - \sum_{i \in I} f_i x_i & \text{profit} = \text{revenue} - \text{cost} \\ \text{s.t.} & \sum_{i \in I} l_i y_i \le L & \text{labor hours constraint} \\ & \sum_{i \in I} s_i y_i \le S & \text{cloth constraint} \\ & y_i \le M_i x_i, \quad \forall i \in I & \text{where } M_i = \min\{L/l_i, S/s_i\} \\ & x_i \text{ binary}, \quad \forall i \in I & \\ & y_i \text{ integer}, \quad \forall i \in I & \end{array}$$

*Textile Manufacturing

Production (continuous)

Input data on products					
	Shirts	Shorts	Pants	Skirts	Jackets
Labor hours	2	1	6	4	8
Cloth	3	2.5	4	4.5	5.5
Selling price	$35	$40	$65	$70	$110
Variable cost	$20	$10	$25	$30	$35
Fixed cost for equipment	$1,500	$1,200	$1,600	$1,500	$1,600
Production plan, constraints on capacity					
	Shirts	Shorts	Pants	Skirts	Jackets
Produce any?	0	1	0	0	1
Units produced	0.00	965.52	0.00	0.00	379.31
	<=	<=	<=	<=	<=
Logical upper limit	0.00	1800.00	0.00	0.00	500.00
Constraints on resources					
	Resource used		Available		
Labor hours	4000.00	<=	4000		
Cloth	4500.00	<=	4500		
Monetary outputs					
Revenue	$80,345				
Total variable cost	$22,931				
Total fixed cost for equipment	$2,800				
Profit	$54,614				

Production (integer)

Input data on products					
	Shirts	Shorts	Pants	Skirts	Jackets
Labor hours	2	1	6	4	8
Cloth	3	2.5	4	4.5	5.5
Selling price	$35	$40	$65	$70	$110
Variable cost	$20	$10	$25	$30	$35
Fixed cost for equipment	$1,500	$1,200	$1,600	$1,500	$1,600
Production plan, constraints on capacity					
	Shirts	Shorts	Pants	Skirts	Jackets
Produce any?	0	1	0	0	1
Units produced	0.00	966.00	0.00	0.00	379.00
	<=	<=	<=	<=	<=
Logical upper limit	0.00	1800.00	0.00	0.00	500.00
Constraints on resources					
	Resource used		Available		
Labor hours	3998.00	<=	4000		
Cloth	4499.50	<=	4500		
Monetary outputs					
Revenue	$80,330				
Total variable cost	$22,925				
Total fixed cost for equipment	$2,800				
Profit	$54,605				

Example: Service Centers

Service Centers

- Customers in eleven cities
- Service centers in three of these cities
- Estimated annual trips to customers in each city

Table 6.10 Estimated Numbers of Annual Trips to Customers

Boston	Chicago	Dallas	Denver	Los Angeles	Miami	New York	Phoenix	Pittsburgh	San Francisco	Seattle
885	760	1124	708	1224	1152	1560	1222	856	1443	612

- Distances between cities

Table 6.9 Distances for the Service Center Example

	Boston	Chicago	Dallas	Denver	Los Angeles	Miami	New York	Phoenix	Pittsburgh	San Francisco	Seattle
Boston	0	983	1815	1991	3036	1539	213	2664	792	2385	2612
Chicago	983	0	1205	1050	2112	1390	840	1729	457	2212	2052
Dallas	1815	1205	0	801	1425	1332	1604	1027	1237	1765	2404
Denver	1991	1050	801	0	1174	2041	1780	836	1411	1765	1373
Los Angeles	3036	2112	1425	1174	0	2757	2825	398	2456	403	1909
Miami	1539	1390	1332	2041	2757	0	1258	2359	1250	3097	3389
New York	213	840	1604	1780	2825	1258	0	2442	386	3036	2900
Phoenix	2664	1729	1027	836	398	2359	2442	0	2073	800	1482
Pittsburgh	792	457	1237	1411	2456	1250	386	2073	0	2653	2517
San Francisco	2385	2212	1765	1765	403	3097	3036	800	2653	0	817
Seattle	2612	2052	2404	1373	1909	3389	2900	1482	2517	817	0

- Objective: Minimize total annual travel distances

*Service Centers

- What are our decisions?

*Service Centers

- What are our decisions?

- Whether to locate a service center in a city
  - Binary
- Whether to assign a city to a service center
  - Binary

Service Centers

- Data
- Decisions
- Model

Table 6.10 Estimated Numbers of Annual Trips to Customers

Boston	Chicago	Dallas	Denver	Los Angeles	Miami	New York	Phoenix	Pittsburgh	San Francisco	Seattle
885	760	1124	708	1224	1152	1560	1222	856	1443	612

Table 6.9 Distances for the Service Center Example

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Dallas	1815	1205	0	801	1425	1332	1604	1027	1237	1765	2404
Denver	1991	1050	801	0	1174	2041	1780	836	1411	1765	1373
Los Angeles	3036	2112	1425	1174	0	2757	2825	398	2456	403	1909
Miami	1539	1390	1332	2041	2757	0	1258	2359	1250	3097	3389
New York	213	840	1604	1780	2825	1258	0	2442	386	3036	2900
Phoenix	2664	1729	1027	836	398	2359	2442	0	2073	800	1482
Pittsburgh	792	457	1237	1411	2456	1250	386	2073	0	2653	2517
San Francisco	2385	2212	1765	1765	403	3097	3036	800	2653	0	817
Seattle	2612	2052	2404	1373	1909	3389	2900	1482	2517	817	0

*Service Centers

- Data
- Decisions
- Model

$$P = \text{number of service centers}$$
$$d_{ij} = \text{distance from city } i \in I \text{ to city } j \in J$$
$$f_j = \text{demand frequency of city } j \in J$$

Table 6.10 Estimated Numbers of Annual Trips to Customers

Boston	Chicago	Dallas	Denver	Los Angeles	Miami	New York	Phoenix	Pittsburgh	San Francisco	Seattle
885	760	1124	708	1224	1152	1560	1222	856	1443	612

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Dallas	1815	1205	0	801	1425	1332	1604	1027	1237	1765	2404
Denver	1991	1050	801	0	1174	2041	1780	836	1411	1765	1373
Los Angeles	3036	2112	1425	1174	0	2757	2825	398	2456	403	1909
Miami	1539	1390	1332	2041	2757	0	1258	2359	1250	3097	3389
New York	213	840	1604	1780	2825	1258	0	2442	386	3036	2900
Phoenix	2664	1729	1027	836	398	2359	2442	0	2073	800	1482
Pittsburgh	792	457	1237	1411	2456	1250	386	2073	0	2653	2517
San Francisco	2385	2212	1765	1765	403	3097	3036	800	2653	0	817
Seattle	2612	2052	2404	1373	1909	3389	2900	1482	2517	817	0

*Service Centers

- Data

$$P = \text{number of service centers}$$
$$d_{ij} = \text{distance from city } i \in I \text{ to city } j \in J$$
$$f_j = \text{demand frequency of city } j \in J$$

- Decisions

$$x_i = \begin{cases} 1, & \text{locate service center in city } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

$$y_{ij} = \begin{cases} 1, & \text{assign service center } i \in I \text{ to serve city } j \in J, \\ 0, & \text{otherwise.} \end{cases}$$

- Model

*Service Centers

- Data

$$P = \text{number of service centers}$$
$$d_{ij} = \text{distance from city } i \in I \text{ to city } j \in J$$
$$f_j = \text{demand frequency of city } j \in J$$

- Decisions

$$x_i = \begin{cases} 1, & \text{locate service center in city } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

$$y_{ij} = \begin{cases} 1, & \text{assign service center } i \in I \text{ to serve city } j \in J, \\ 0, & \text{otherwise.} \end{cases}$$

- Model

$$\begin{array}{lll} \min & \sum_{j \in J} f_j \sum_{i \in I} d_{ij} y_{ij} & \text{minimize total distance} \\ \text{s.t.} & \sum_{i \in I} x_i \leq P & \text{number of service centers} \\ & \sum_{i \in I} y_{ij} = 1, \quad \forall j \in J & \text{assign one service center to each city } j \in J \\ & y_{ij} \leq x_i, \quad \forall i \in I, j \in J & \text{city } i \text{ can serve city } j \text{ only if city } i \text{ is chosen to be a service center} \\ & x_i \text{ binary}, \quad \forall i \in I & \\ & y_{ij} \text{ binary}, \quad \forall i \in I, j \in J & \end{array}$$

*Service Centers

Distances between cities

	Boston	Chicago	Dallas	Denver	Los Angeles	Miami	New York	Phoenix	Pittsburgh	San Francisco	Seattle
Boston	0	983	1815	1991	3036	1539	213	2664	792	2385	2612
Chicago	983	0	1205	1050	2112	1390	840	1729	457	2212	2052
Dallas	1815	1205	0	801	1425	1332	1604	1027	1237	1765	2404
Denver	1991	1050	801	0	1174	2041	1780	836	1411	1765	1373
Los Angeles	3036	2112	1425	1174	0	2757	2825	398	2456	403	1909
Miami	1539	1390	1332	2041	2757	0	1258	2359	1250	3097	3389
New York	213	840	1604	1780	2825	1258	0	2442	386	3036	2900
Phoenix	2664	1729	1027	836	398	2359	2442	0	2073	800	1482
Pittsburgh	792	457	1237	1411	2456	1250	386	2073	0	2653	2517
San Francisco	2385	2212	1765	1765	403	3097	3036	800	2653	0	817
Seattle	2612	2052	2404	1373	1909	3389	2900	1482	2517	817	0

Locations of service centers

	Boston	Chicago	Dallas	Denver	Los Angeles	Miami	New York	Phoenix	Pittsburgh	San Francisco	Seattle	Total service centers		Allowed
Included?	0	0	1	0	0	0	1	0	0	1	0	3	<=	3

Assignments (1 if customers along side are serviced by service center along top, 0 otherwise)

	Boston	Chicago	Dallas	Denver	Los Angeles	Miami	New York	Phoenix	Pittsburgh	San Francisco	Seattle	Locations assigned to		Required
Boston	0	0	0	0	0	0	1	0	0	0	0	1	=	1
Chicago	0	0	0	0	0	0	1	0	0	0	0	1	=	1
Dallas	0	0	1	0	0	0	0	0	0	0	0	1	=	1
Denver	0	0	1	0	0	0	0	0	0	0	0	1	=	1
Los Angeles	0	0	0	0	0	0	0	0	0	1	0	1	=	1
Miami	0	0	0	0	0	0	1	0	0	0	0	1	=	1
New York	0	0	0	0	0	0	1	0	0	0	0	1	=	1
Phoenix	0	0	0	0	0	0	0	0	0	1	0	1	=	1
Pittsburgh	0	0	0	0	0	0	1	0	0	0	0	1	=	1
San Francisco	0	0	0	0	0	0	0	0	0	1	0	1	=	1
Seattle	0	0	0	0	0	0	0	0	0	1	0	1	=	1

Copies of "Included?" row

Boston	0	0	1	0	0	0	1	0	0	1	0
Chicago	0	0	1	0	0	0	1	0	0	1	0
Dallas	0	0	1	0	0	0	1	0	0	1	0
Denver	0	0	1	0	0	0	1	0	0	1	0
Los Angeles	0	0	1	0	0	0	1	0	0	1	0
Miami	0	0	1	0	0	0	1	0	0	1	0
New York	0	0	1	0	0	0	1	0	0	1	0
Phoenix	0	0	1	0	0	0	1	0	0	1	0
Pittsburgh	0	0	1	0	0	0	1	0	0	1	0
San Francisco	0	0	1	0	0	0	1	0	0	1	0
Seattle	0	0	1	0	0	0	1	0	0	1	0

Numbers of annual trips to customers, and total distances (1000s of miles) traveled annually to customers

	Annual trips	Total annual distance
Boston	885	189
Chicago	760	638
Dallas	1124	0
Denver	708	567
Los Angeles	1224	493
Miami	1152	1449
New York	1560	0
Phoenix	1222	978
Pittsburgh	856	330
San Francisco	1443	0
Seattle	612	500

Objective to minimize (1000s of miles)

Total distance	5,145

*Service Centers

- Data

$$P = \text{number of service centers}$$
$$d_{ij} = \text{distance from city } i \in I \text{ to city } j \in J$$
$$f_j = \text{demand frequency of city } j \in J$$

- Decisions

$$x_i = \begin{cases} 1, & \text{locate service center in city } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

$$y_{ij} = \begin{cases} 1, & \text{assign service center } i \in I \text{ to serve city } j \in J, \\ 0, & \text{otherwise.} \end{cases}$$

- Model

$$\begin{array}{lll}
\min & \sum_{j \in J} f_j \sum_{i \in I} d_{ij} y_{ij} & \text{minimize total distance} \\
\text{s.t.} & \sum_{i \in I} x_i \leq P & \text{number of service centers} \\
& \sum_{i \in I} y_{ij} = 1, \quad \forall j \in J & \text{assign one service center to each city } j \in J \\
& \sum_{j \in J} y_{ij} \leq M x_i, \quad \forall i \in I & \text{city } i \text{ can serve other cities only if it is chosen to be a service center} \\
& x_i \text{ binary}, \quad \forall i \in I & \\
& y_{ij} \text{ binary}, \quad \forall i \in I, j \in J &
\end{array}$$

*Service Centers

Distances between cities

	Boston	Chicago	Dallas	Denver	Los Angeles	Miami	New York	Phoenix	Pittsburgh	San Francisco	Seattle
Boston	0	983	1815	1991	3036	1539	213	2664	792	2385	2612
Chicago	983	0	1205	1050	2112	1390	840	1729	457	2212	2052
Dallas	1815	1205	0	801	1425	1332	1604	1027	1237	1765	2404
Denver	1991	1050	801	0	1174	2041	1780	836	1411	1765	1373
Los Angeles	3036	2112	1425	1174	0	2757	2825	398	2456	403	1909
Miami	1539	1390	1332	2041	2757	0	1258	2359	1250	3097	3389
New York	213	840	1604	1780	2825	1258	0	2442	386	3036	2900
Phoenix	2664	1729	1027	836	398	2359	2442	0	2073	800	1482
Pittsburgh	792	457	1237	1411	2456	1250	386	2073	0	2653	2517
San Francisco	2385	2212	1765	1765	403	3097	3036	800	2653	0	817
Seattle	2612	2052	2404	1373	1909	3389	2900	1482	2517	817	0

Locations of service centers

	Boston	Chicago	Dallas	Denver	Los Angeles	Miami	New York	Phoenix	Pittsburgh	San Francisco	Seattle	Total service centers		Allowed
Included?	0	0	1	0	0	0	1	0	0	1	0	3	<=	3

Assignments (1 if customers along side are serviced by service center along top, 0 otherwise)

	Boston	Chicago	Dallas	Denver	Los Angeles	Miami	New York	Phoenix	Pittsburgh	San Francisco	Seattle	Locations assigned to		Required
Boston	0	0	0	0	0	0	1	0	0	0	0	1	=	1
Chicago	0	0	0	0	0	0	1	0	0	0	0	1	=	1
Dallas	0	0	1	0	0	0	0	0	0	0	0	1	=	1
Denver	0	0	1	0	0	0	0	0	0	0	0	1	=	1
Los Angeles	0	0	0	0	0	0	0	0	0	1	0	1	=	1
Miami	0	0	0	0	0	0	1	0	0	0	0	1	=	1
New York	0	0	0	0	0	0	1	0	0	0	0	1	=	1
Phoenix	0	0	0	0	0	0	0	0	0	1	0	1	=	1
Pittsburgh	0	0	0	0	0	0	1	0	0	0	0	1	=	1
San Francisco	0	0	0	0	0	0	0	0	0	1	0	1	=	1
Seattle	0	0	0	0	0	0	0	0	0	1	0	1	=	1
Aggregate assignments	0	0	2	0	0	0	5	0	0	4	0			
Big M	11													
Service capacity	0	0	11	0	0	0	11	0	0	11	0			

Numbers of annual trips to customers, and total distances (1000s of miles) traveled annually to customers

	Annual trips	Total annual distance
Boston	885	189
Chicago	760	638
Dallas	1124	0
Denver	708	567
Los Angeles	1224	493
Miami	1152	1449
New York	1560	0
Phoenix	1222	978
Pittsburgh	856	330
San Francisco	1443	0
Seattle	612	500

Objective to minimize (1000s of miles)

Total distance	5,145

IP Modeling Techniques

IP Modeling Techniques

- Fixed charge model

$$f_i(x_i) = \begin{cases} k_i + c_i x_i, & \text{if } x_i > 0, \\ 0, & \text{if } x_i = 0. \end{cases}$$

*IP Modeling Techniques

Fixed charge model

$$f_i(x_i) = \begin{cases} k_i + c_i x_i, & \text{if } x_i > 0, \\ 0, & \text{if } x_i = 0. \end{cases}$$

$$\boxed{\begin{array}{lll} \min & \sum_i (k_i y_i + c_i x_i) & \\ \text{s.t.} & x_i \le M y_i, & \forall i, \\ & y_i \text{ binary}, & \forall i, \\ & x_i \ge 0, & \forall i. \end{array}}$$

- Example: Textile manufacturing

$$\boxed{\begin{array}{lll} \max & \sum_i p_i y_i - \sum_i c_i y_i - \sum_i f_i x_i & \text{profit} = \text{revenue} - \text{cost} \\ \text{s.t.} & \sum_i l_i y_i \le L & \text{labor hours constraint} \\ & \sum_i s_i y_i \le S & \text{cloth constraint} \\ & y_i \le M_i x_i, \quad \forall i & \text{where } M_i = \min\{L/l_i, S/s_i\} \\ & x_i \text{ binary}, \quad \forall i & \\ & y_i \ge 0, \quad \forall i & \end{array}}$$

IP Modeling Techniques

- Production mix
  - Produce at most K out of N types of products

*IP Modeling Techniques

- Production mix
  - Produce at most K out of N types of products

$$\boxed{\begin{array}{lll} & x_i & \leq My_i, \quad \forall i, \\ \sum_i^N & y_i & \leq K, \\ & y_i & \text{binary}, \quad \forall i, \\ & x_i & \geq 0, \quad \forall i. \end{array}}$$

IP Modeling Techniques

- Forcing constraints
  - Decision A (e.g., $x = 1$) can be made only if decision B (e.g., $y = 1$) has also been made.

*IP Modeling Techniques

- Forcing constraints
  - Decision A (e.g., $x = 1$) can be made only if decision B (e.g., $y = 1$) has also been made.

$$\Rightarrow \quad \boxed{\begin{array}{ll} x & \leq y, \\ x, y & \text{binary.} \end{array}}$$

  - Example: Service center (P-median) problem

$$\boxed{\begin{array}{lll} \min & \sum_j f_j \sum_i d_{ij} y_{ij} & \text{minimize total distance} \\ \text{s.t.} & \sum_i x_i \leq P & \text{number of service centers} \\ & \sum_i y_{ij} = 1, \quad \forall j & \text{assign one service center to each city} \\ & y_{ij} \leq x_i, \quad \forall i, j & \text{a city can serve another city only if it is chosen to be a service center} \\ & x_i \text{ binary}, \quad \forall i & \\ & y_{ij} \text{ binary}, \quad \forall i, j & \end{array}}$$

IP Modeling Techniques

- Disjunctive constraints

$$\begin{aligned} ax &\leq b, \\ &\text{or} \\ cx &\leq d. \end{aligned}$$

  - Example

$$\begin{aligned} 3x_1 + 2x_2 &\leq 18 \\ &\text{or} \\ x_1 + 4x_2 &\leq 16 \end{aligned}$$

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IP Modeling Techniques

- Disjunctive constraints (at least K of them are satisfied)

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\boxed{
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&\dots \\
a_N x &\leq b_N + M y_N, \\
\textstyle\sum_{i=1}^{N} y_i &\leq N - K, \\
y_i &\text{ binary}, \forall i.
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}
$$

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- Restricted range of values
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IP Modeling Techniques

- Minimize a piecewise linear (PWL) cost function
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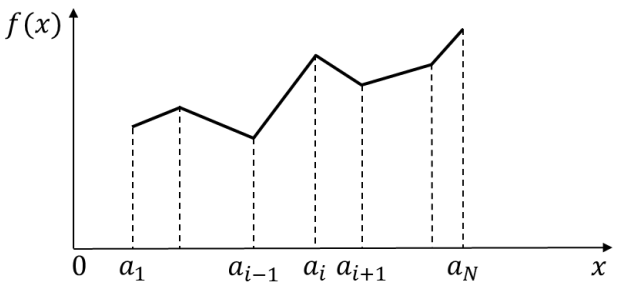

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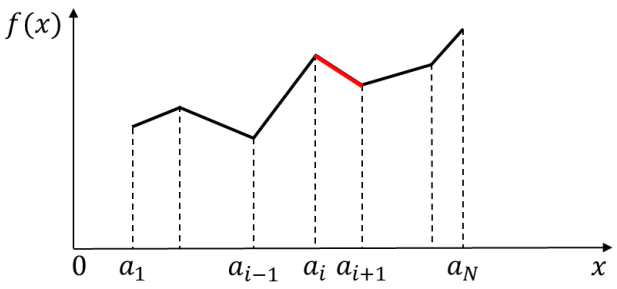

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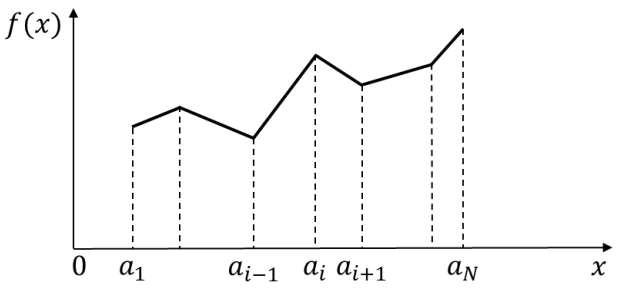

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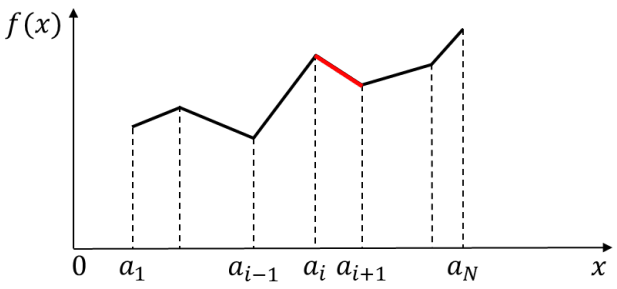

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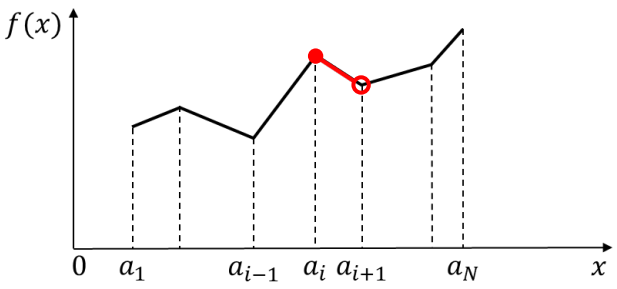

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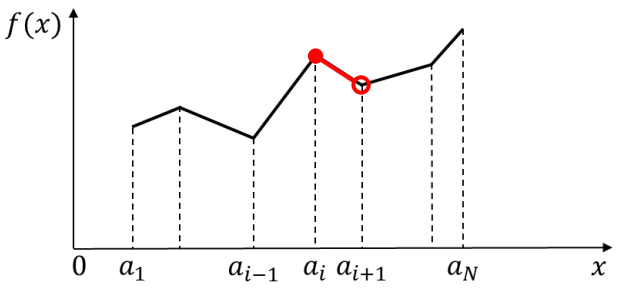

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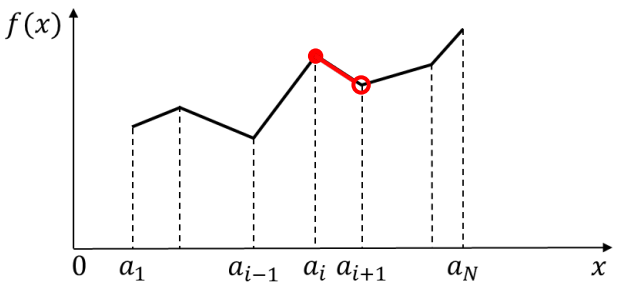

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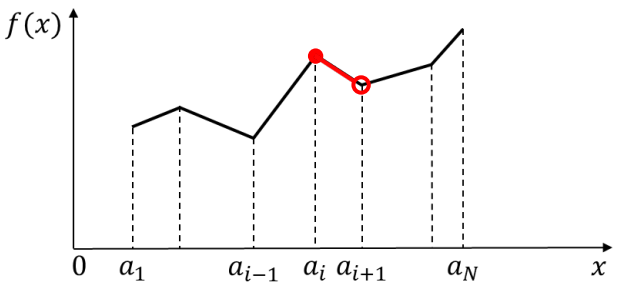

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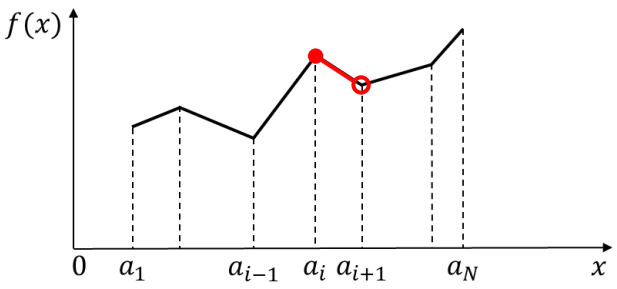

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 & \lambda_1 \le y_1, \quad \lambda_N \le y_{N-1}, & \text{boundary constraints} \\
 & \lambda_i \le y_{i-1} + y_i, \quad i = 2,\dots,N-1, & \text{at most two consecutive } \lambda_i canbenonzero \\
 & & \text{i.e., } y_i = 1 \Rightarrow \text{only } \lambda_i \text{ and } \lambda_{i+1} \text{ can be nonzero} \\
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Q & A